JOC The Journal of Organic Chemistry

pubs.acs.org/joc Note

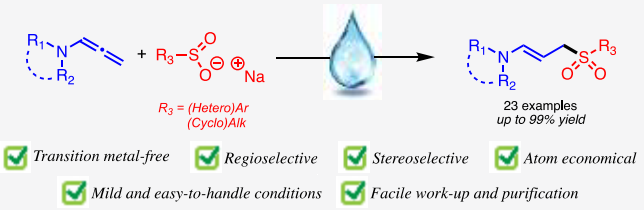
Easy Access to Allylic Sulfones Through Transition-Metal-Free Hydrosulfonylation Of Allenes

Lucas Pagès, Sébastien Lemouzy, Marc Taillefer,* and Florian Monnier*

Cite This: J. Org. Chem. 2021, 86, 15695−15701 Read Online

ACCESS | Metrics & More | Article Recommendations | Supporting Information

ABSTRACT: A Brønsted acid-mediated addition of (hetero)aryl and (cyclo)alkyl sodium sulfinates to *N*-allenyl derivatives, which proceeds in water, is described under very smooth conditions. This reaction provided a practical and efficient protocol for the regio- and stereoselective synthesis of allylic sulfones in an atom- and step-economic fashion.

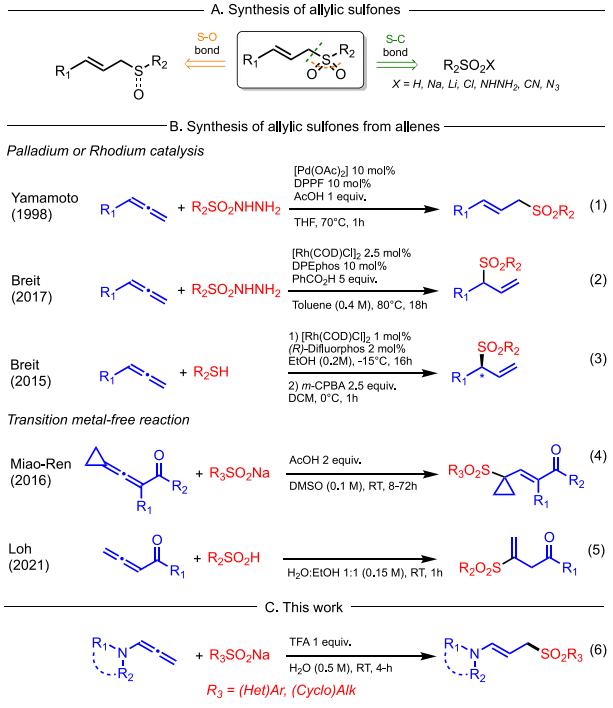


Downloaded via FUDAN UNIV on January 17, 2022 at 11:23:06 (UTC).
See https://pubs.acs.org/sharingguidelines for options on how to legitimately share published articles.

Allylic sulfones are highly valuable structures commonly found in bioactive molecules for pharmaceutical or agrochemical use.[1−7] They can also be useful intermediates toward various functionalized compounds, either through the C(sp^2)−C(sp^2) bond or the carbon atom adjacent to the sulfone group.[8−10] The SO_2 moiety is classically introduced either through the in situ oxidation of sulfides (S^{II}) or sulfoxides (S^{IV}) or the use of preoxidized sulfur-based reagents (S^{VI}), i.e., sulfinic acid derivatives and their salts, in substitution or cross-coupling reactions (Scheme 1A).

Allenes are interesting precursors of these allylic structures that are accessed via hydrofunctionalization[11] with either hydrothiolation followed by in situ oxidation or direct hydrosulfonylation. The first allene hydrosulfonylation strategies were based on the use of sulfonylhydrazines as a source of sulfone. They can be catalyzed by a palladium and diphosphine system (Yamamoto,[12] Scheme 1, eq 1) or by a rhodium and DPEPhos system (Breit,[13] Scheme 1, eq 2), occurring on the distal carbon or the proximal carbon of the allene, respectively. These methods require the presence of a Brønsted acid in at least a stoichiometric quantity and the use of an organic solvent under a moderately high temperature (70−80 °C). Breit's group also developed an asymmetric rhodium-catalyzed hydrothiolation of terminal allenes, followed by oxidation, to afford the chiral branched allylic sulfones in a two-step process (Scheme 1, eq 3).[14] Over the last years, two transition-metal-free allene hydrosulfonylation methods have been described. In 2016, Miao, Ren, and coworkers reported a transition-metal-free tunable hydrosulfonylation of 3-cyclopropylideneprop-2-en-1-ones using sodium sulfinates in the presence of acetic acid, which afforded the corresponding allylic sulfone (γ-adduct) in DMSO[15,16] (Scheme 1, eq 4). Very recently, the group of Loh reported a transition-metal-catalyst-free regioselective Michael addition of sulfinic acids to α-ketoallenes in a mixture of water and ethanol as a solvent, which afforded the corresponding vinylic sulfone[17] (Scheme 1, eq 5). This method requires the synthesis

Scheme 1. Metal-Catalyzed or Metal-Free Hydrosulfonylation of Allenes: Known Methods and Novel Pathway Described Herein with TFA



Received: June 8, 2021
Published: October 18, 2021



© 2021 American Chemical Society

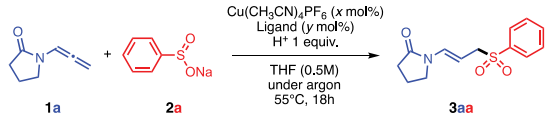
https://doi.org/10.1021/acs.joc.1c01345
J. Org. Chem. 2021, 86, 15695−15701

of sulfinic acids in one or two steps from either sodium sulfinate or sulfonyl chloride.[18]

In our course to build new original allylic structures, we started to investigate C−S bond formation via the copper-catalyzed hydrofunctionalization of allenes. After starting with the usual conditions described in our previous work for the C−C,[19−21] C−N,[22−25] and C−O[26] bond formation, we quickly realized that the reaction could occur without a transition metal in the presence of a Brønsted acid and under very smooth and easy-to-handle conditions. We herein describe an efficient synthesis of allylic sulfones and 1,3-disulfones through the regio- and stereoselective single and double addition of various (hetero)aryl and (cyclo)alkyl sodium sulfinates to *N*-allenyl derivatives in aqueous conditions at room temperature in the presence of trifluoroacetic acid (TFA).

Based on our previous works, we began to study the model reaction between *N*-allenyl-2-pyrrolidinone **1a** and the benzene sulfinic acid sodium salt **2a** under our Cu-$(CH_3CN)_4PF_6$-catalyzed system (Table 1). With an acid

Table 1. Copper-Catalyzed Hydrosulfonylation of 1a with 2a: Selected Data for Optimization of the Reaction Conditions[a]



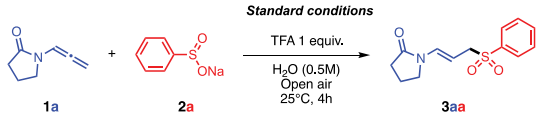
entry	*x*	*y*	ligand[d]	H^+ source	conversion (%)[b,c]	yield (%)[c]
1	0	0	none	none	0	0
2	10	0	none	*p*-toluic acid	63%	50%
3	10	10	Bpy	*p*-toluic acid	86%	57%
4	10	10	Phen	*p*-toluic acid	74%	33%
5	10	10	TMEDA	*p*-toluic acid	95%	51%
6	10	10	Bpy	TMBA	79%	39%
7	10	10	Bpy	PTSA	76%	30%
8	10	10	Bpy	TFA	>99%	58%
9	0	0		TFA	>99%	90%

[a]Reaction conditions are as follows: **1a** (0.25 mmol), **2a** (2 equiv, 0.5 mmol), H^+ source (1 equiv). The catalyst and the ligand were placed in a Schlenk tube under argon in 0.5 mL of THF (0.5M) for 18 h at 55 °C. [b]Based on the recovered starting material **1a**. [c]Determined through ^{1}H NMR analysis using trichloroethylene (1 equiv, 0.25 mmol) as an internal standard. [d]TMBA = 2,4,6-trimethylbenzoic acid. PTSA = *p*-toluenesulfonic acid. Bpy = 2,2′-bipyridine. Phen = 1,10-phenantroline. TMEDA = *N*,*N*,*N*′,*N*′-tetramethylethylenediamine.

source, we showed that the presence of a catalytic amount of copper allows the formation of a 50% yield of **3aa** (entry 2, Table 1). Different ligands were associated with copper, but no real improvement was detected (entries 3−5, Table 1). In the next set of experiments, we tested different acid sources (entries 6−8, Table 1) and defined trifluoroacetic acid (TFA) as the best one. However, we surprisingly observed that **3aa** could be obtained in a total conversion with a 90% yield in the absence of the copper system (entry 9, Table 1).

As mentioned above in Table 1, the best conditions were based on the use of only 1 equiv of TFA in the absence of the copper catalyst. Hence, we opted to explore milder conditions (Table 2) and were finally able to perform the reaction in water, under open air, and at room temperature. Thanks to these improvements, the desired product **3aa** was obtained in a quantitative yield.

Table 2. Acid-Mediated Hydrosulfonylation of 1a with 2a: Selected Data for the Reaction Parametric Study[a]

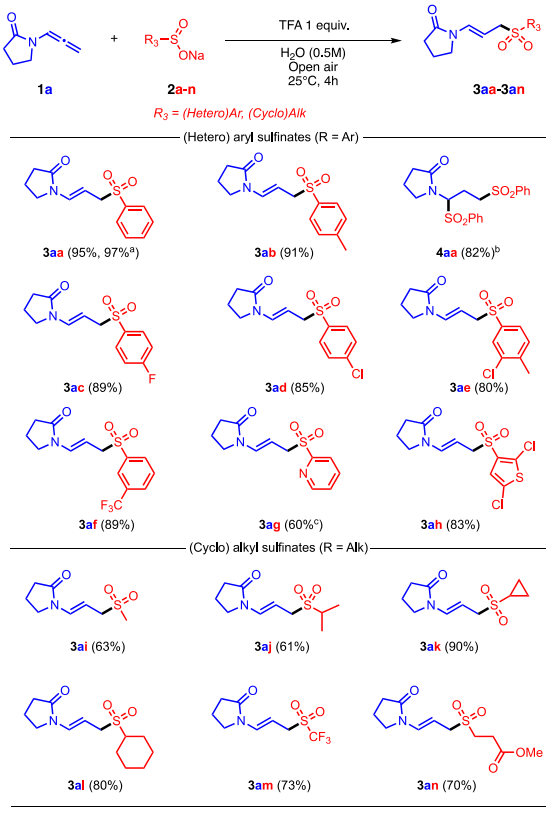


entry	variations from the standard conditions	conversion (%)[b,c]	yield (%)[c]
1	THF instead of H_2O	>99%	80%
2	under argon instead of open air	>99%	90%
3	55 °C instead of 25 °C	>99%	96%
4	no TFA	66%	0%
5	none	>99%	>99%

[a]Reaction conditions are as follows: **1a** (0.25 mmol), **2a** (2 equiv, 0.5 mmol), and TFA (1 equiv, 0.25 mmol) were placed in a tube under open air in 0.5 mL of solvent (0.5M) for 4 h at 25 °C. [b]Based on the recovered starting material. [c]Determined through ^{1}H NMR analysis using trichloroethylene (1 equiv, 0.25 mmol) as an internal standard.

With these optimized conditions in hand, we first explored the scope of the method for the hydrosulfonylation of *N*-allenyl-2-pyrrolidinone **1a** with various readily available (hetero)aryl and (cyclo)alkyl sodium sulfinates compounds **2a−2n** (Scheme 2). Different *para*-substituted benzene

Scheme 2. TFA-Mediated Hydrosulfonylation of Allene 1a with Various Sodium Sulfinates 2a−2n[d]



[a]The reaction was performed on a 2 mmol scale. [b]The reaction was performed at 40 °C over 2 days under open air with 4 equiv of TFA, 4 equiv of $PhSO_2Na$, and H_2O/THF 1:1 (0.25 M) as the solvent. [c]The reaction was performed with 2 equiv of TFA. [d]Reaction conditions are follows: **1a** (0.25 mmol), **2a−n** (2 equiv, 0.5 mmol), TFA (1 equiv, 0.25 mmol), H_2O (0.5 M, 0.5 mL), open air, 25 °C, 4 h. Isolated yields.

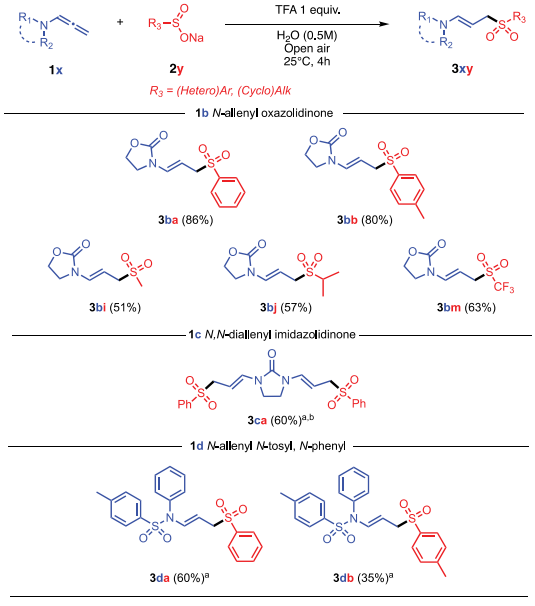
https://doi.org/10.1021/acs.joc.1c01345
J. Org. Chem. 2021, 86, 15695−15701

sodium sulfinates were tested, such as sodium *p*-toluenesulfinate **2b** and 4-fluoro- and 4-chlorobenzenesulfinic acids **2c** and **2d**, which afforded the corresponding products **3ab**−**3ad** in high yields (85−91%). *meta*-Substituted aryl sulfinate partners such as the 3-chloro-4-methylbenzenesulfinic acid sodium salt **2e** and the 3-(trifluoromethyl)benzenesulfinic acid sodium salt **2f** were also successfully engaged in the reaction and gave the corresponding products in 80−89% yields. Finally, *N*- and *S*-heteroaromatics **2g** and **2h**, respectively, provided the desired products **3ag** and **3ah** in good to high yields (60−83%). Alkyl sulfinates were then investigated, and both linear (**2i** and **2j**) and cyclic (**2k** and **2l**) derivatives yielded the hydrosulfonylation products in good isolated yields (61−75%). Functionalized sulfinates such as sodium triflinate **2m** and sodium 1-methyl-3-sulfinopropanoate **2n** were also able to provide the allylic sulfones in good yields (70−73%), showing the high tolerance of the process. To prove the robustness of the reaction, we were interested in performing it on a larger scale. The reaction between our model substrates **1a** and **2a** was then conducted on a 2 mmol scale, i.e., on a scale eight times larger than usual, and afforded the corresponding hydrosulfonylation product **3aa** with a 97% isolated yield.

Interestingly, we also performed a one-step double hydrosulfonylation of **1a** with **2a** to obtain the corresponding 1,3-disulfone **4aa** in a high yield. It is worth mentioning that the same 1,3-disulfone **4aa** can also be obtained through the hydrosulfonylation of **3aa**, demonstrating that our reaction can be applied both to *N*-allenyl and *N*-alkenyl derivatives.

We then explored the reactivity of different *N*-allenyl derivatives with several (hetero)aryl and (cyclo)alkyl sodium sulfinates (Scheme 3). The *N*-allenyl-2-oxazolidinone **1b** was

Scheme 3. TFA-Mediated Hydrosulfonylation of Various Allenyl Derivatives with Different Sodium Sulfinates[c]



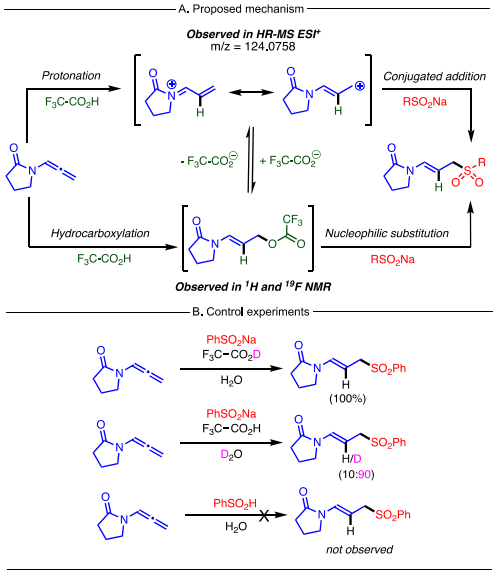
[a]H_2O/THF 1:1 (0.5 M) was used as the solvent. [b]The reaction was performed using 2 equiv of TFA and 4 equiv of $PhSO_2Na$. [c]Reaction conditions (unless otherwise mentioned) are as follows: **1** (0.25 mmol), **2** (2 equiv, 0.5 mmol), TFA (1 equiv, 0.25 mmol), H_2O (0.5 M, 0.5 mL), open air, 25 °C, 4 h. Isolated yields.

successfully involved in the reaction, affording both aryl and alkyl allylic sulfones in medium to excellent yields (51−90%).

The symmetrical diallene *N*,*N*-diallenyl-2-imidazolidinone **1c** afforded the corresponding double hydrosulfonylation product **3ca** in a 60% yield when a mixture 1:1 of water and THF was used as the solvent. We also investigated the reactivity of *N*-sulfonyl allenamides and observed that these substrates and their corresponding hydrofunctionalization products were sensitive under our acidic conditions. Nevertheless, the sulfones **3da** and **3db** were obtained in moderate to good isolated yields (35−60%).

The mechanism of the reaction could take place in two stages. The first step is the hydrocarboxylation of the allene with TFA that leads to the formation of the corresponding allylic ester (observed with both ^1H and ^{19}F NMR), which is in equilibrium with the corresponding iminium and carbocation forms in the presence of water (observed with HRMS ESI+). The second step could be either a nucleophilic substitution on the ester by the sulfinate with the trifluoroacetate as a leaving group or a conjugated addition of the sulfinate to the iminium species, which finally leads to the formation of the desired allylic sulfone. The nucleophilic addition occurred on the *β*-position of the iminium salt, which is a classical pathway to obtain enamines. (Scheme 4A.).[27] When the reaction was

Scheme 4. Proposed Mechanism and Control Experiments



performed using D_2O as a solvent in the presence of either TFA or deuterated TFA, 90% deuterium incorporation was observed at the central position of the allene, which is consistent with the mechanism proposed above. No product was formed when benzenesulfinic acid ($PhSO_2H$) was used instead of the corresponding sodium salt ($PhSO_2Na$), suggesting that the sulfinic acid is not strong enough to protonate the *N*-allenyl derivative. (Scheme 4B).

To conclude, we developed a Brønsted acid-mediated allylation of S(VI)-containing nucleophiles. This transition-metal-free methodology occurs in mild conditions and allows the hydrosulfonylation of a wide range of *N*-allenyl derivatives substrates with total regio- and stereoselectivity in water. Unprecedented (*E*)-allylic sulfones and a 1,3-disulfone were obtained with moderate to excellent yields in a completely atom-economical fashion. Further investigations and mechanistic studies for the hydrofunctionalization of allenes will be reported in due course.

https://doi.org/10.1021/acs.joc.1c01345
J. Org. Chem. 2021, 86, 15695−15701

EXPERIMENTAL SECTION

General Considerations. All reactions were performed in oven-dried Schlenk flasks under open air. Unless otherwise mentioned, all reagents were purchased from commercial sources and were used without further purification and weighed to air. Allenes were either purchased from commercial sources or synthesized according to methods previously described. HPLC-grade water was systematically used as solvent. The details of the experimental procedures and characterization data (^{1}H, ^{13}C, and ^{19}F NMR; ESI-HRMS; and m.p., if solid) of all the new compounds are in the Supporting Information. ^{1}H, ^{1}H{^{19}F}, ^{13}C{^{1}H}, and ^{19}F{^{1}H} NMR spectra were recorded with a Bruker AC-400 MHz spectrometer in $CDCl_3$, and the residual solvent protons (7.26 for ^{1}H) or carbons (77.16 for ^{13}C) were used as internal references. Chemical shifts (δ) are reported in parts per million (ppm), and the coupling constants (J) are reported in Hertz (Hz). The following abbreviations are used: *s*, singlet; *d*, doublet; *dd*, doublet of doublets; *dt*, doublet of triplets; *t*, triplet; *hept*, heptuplet; and *m*, multiplet.

A TOF-type mass analyzer was used for the HRMS measurements. Electrospray ionization (ESI) high-resolution mass spectra were recorded on a Waters SYNAPT G2-S (SN: UEB205) high-definition mass spectrometer in the positive ion mode from 100 to 1500 Da. Products were dissolved in either MeOH or a basic aqueous solution, depending on the case, and were introduced directly into the spectrometer. The capillary voltage was 3000 V, and the cone voltage was 30 V. The source and desolvation temperatures were 100 and 150 °C respectively. The data were reprocessed by the Masslynx 4.1 software. Melting points (m.p.) were manually determined in degrees centigrade (°C) with a Buchi Melting Point M-560 apparatus.

General Procedure. In a tube of an appropriate size was added under open air sodium sulfinate **2** (2 equiv, 0.5 mmol), solvent (0.5 M, 0.5 mL), allene **1** (1 equiv, 0.25 mmol), and trifluoroacetic acid (TFA) (1 eq, 0.25 mmol). The mixture was stirred at 25 °C for 4 h under open air unless otherwise mentioned. After an aqueous workup, the organic phase was separated. The remaining aqueous layer was further extracted with DCM. The gathered organic layers were then dried over anhydrous magnesium sulfate and concentrated under vacuum. Then, trichloroethylene (1 equiv, 0.25 mmol) was added as the internal standard to estimate the NMR yield. In most cases, the analytically pure hydrosulfonylation product could be obtained directly without further purification; however, purification by triethylamine-treated silica gel column chromatography was used if necessary .

General Procedure for Large Scale Synthesis. In a tube of appropriate size was added under open air sodium sulfinate **2** (2 equiv, 4 mmol), solvent (0.5 M, 4 mL), allene **1** (1 equiv, 2 mmol), and trifluoroacetic acid (TFA) (1 eq, 2 mmol). The mixture was stirred at 25 °C for 4 h under open air. After an aqueous workup, the organic phase was separated. The remaining aqueous layer was further extracted with DCM. The gathered organic layers were then dried over anhydrous magnesium sulfate and concentrated under vacuum.

3aa *(E)-1-(3-(Phenylsulfonyl)prop-1-en-1-yl)pyrrolidin-2-one* $C_{13}H_{15}NO_3S$. The reaction of *N*-allenyl-2-pyrrolidinone (**1a**, 0.25 mmol) with benzenesulfinic acid sodium salt (**2a**, 2 equiv, 0.5 mmol) afforded the corresponding hydrosulfonylation product **3aa** as a white solid (63.0 mg, 95%) according to the general procedure. The product **3aa** was obtained in a 97% yield (514.7 mg) according to the general procedure for the large-scale synthesis. ^{1}H NMR (400 MHz, $CDCl_3$): δ 7.89−7.84 (*m*, 2H), 7.68−7.63 (*m*, 1H), 7.58−7.53 (*m*, 2H), 6.87 (*d*, J = 14.4 Hz, 1H), 4.89 (*dt*, J = 14.4, 7.7 Hz, 1H), 3.81 (*dd*, J = 7.7, 1.1 Hz, 2H), 3.55−3.48 (*m*, 2H), 2.50−2.44 (*m*, 2H), 2.11 (*m*, 2H). ^{13}C{^{1}H} NMR (101 MHz, $CDCl_3$): δ 173.3, 138.4, 133.9, 131.1, 129.2, 128.4, 96.4, 58.6, 45.0, 30.9, 17.4. HRMS (ESI) m/z: $[M + H]^+$ Calcd for $C_{13}H_{16}NO_3S$ 266.0845, found 266.9845. $[M + Na]^+$ Calcd for $C_{13}H_{15}NO_3SNa$ 288.0665, found 288.0683. m.p.: 170−172 °C

3ab *(E)-1-(3-Tosylprop-1-en-1-yl)pyrrolidin-2-one* $C_{14}H_{17}NO_3S$. The reaction of *N*-allenyl-2-pyrrolidinone (**1a**, 0.25 mmol) with *p*-toluenesulfinic acid sodium salt (**2b**, 2 equiv, 0.5 mmol) afforded the corresponding hydrosulfonylation product **3ab** as a white solid (63.6 mg, 91%) according to the general procedure. ^{1}H NMR (400 MHz, $CDCl_3$): δ 7.76−7.71 (*m*, 2H), 7.36−7.32 (*m*, 2H), 6.87 (*d*, J = 14.4 Hz, 1H), 4.90 (*dt*, J = 14.4, 7.7 Hz, 1H), 3.78 (*dd*, J = 7.7, 1.1 Hz, 2H), 3.51 (*dd*, J = 8.8, 5.6 Hz, 2H), 2.47 (*dd*, J = 8.7, 7.6 Hz, 2H), 2.44 (*s*, 3H), 2.17−2.08 (*m*, 2H). ^{13}C{^{1}H} (101 MHz, $CDCl_3$): δ 173.3, 144.9, 135.5, 130.9, 129.8, 128.4, 96.7, 58.6, 45.0, 30.96, 21.68, 17.39. HRMS (ESI) m/z:$[M + H]^+$ Calcd for $C_{14}H_{18}NO_3S$ 280.1002, found 280.1009. $[M + Na]^+$ Calcd for $C_{14}H_{17}NO_3SNa$ 302.0821, found 302.0825. m.p.: 138−142 °C

3ac *(E)-1-(3-((4-Fluorophenyl)sulfonyl)prop-1-en-1-yl)pyrrolidin-2-one* $C_{13}H_{14}FNO_3S$. The reaction of *N*-allenyl-2-pyrrolidinone (**1a**, 0.25 mmol) with 4-fluorobenzene sulfinic acid sodium salt (**2c**, 2 equiv, 0.5 mmol) afforded the corresponding hydrosulfonylation product **3ac** as a white solid (63.0 mg, 89%) according to the general procedure. ^{1}H{^{19}F} NMR (400 MHz, $CDCl_3$): δ 7.90−7.85 (*m*, 2H), 7.25−7.21 (*m*, 2H), 6.89 (*d*, J = 14.4 Hz, 1H), 4.88 (*dt*, J = 14.4, 7.8 Hz, 1H), 3.81 (*dd*, J = 7.7, 1.1 Hz, 2H), 3.55−3.47 (*m*, 2H), 2.48 (*t*, J = 8.2 Hz, 2H), 2.18−2.08 (*m*, 2H). ^{13}C{^{1}H} NMR (101 MHz, $CDCl_3$): δ 173.4 (*s*), 165.9 (*d*, J = 256.6 Hz), 134.5 (*d*, J = 3.3 Hz), 131.2 (*d*, J = 9.6 Hz), 131.2 (*s*), 116.6 (*d*, J = 22.6 Hz), 96.2 (*s*), 58.7 (*s*), 45.0 (*s*), 30.9 (*s*), 17.4 (*s*). ^{19}F{^{1}H} NMR (377 MHz, $CDCl_3$): δ −103.06 (*s*). HRMS (ESI) m/z: $[M + H]^+$ Calcd for $C_{13}H_{15}FNO_3S$.284.0751, found 284.0758. $[M + Na]^+$ Calcd for $C_{13}H_{14}FNO_3SNa$.306.0571, found 306.0570. m.p.: 196−198 °C

3ad *(E)-1-(3-((4-Chlorophenyl)sulfonyl)prop-1-en-1-yl)-pyrrolidin-2-one* $C_{13}H_{14}ClNO_3S$. The reaction of *N*-allenyl-2-pyrrolidinone (**1a**, 0.25 mmol) with 4-chlorobenzene sulfinic acid sodium salt (**2d**, 2 equiv, 0.5 mmol) afforded the corresponding hydrosulfonylation product **3ad** as a white solid (63.7 mg, 85%) according to the general procedure. ^{1}H NMR (400 MHz, $CDCl_3$): δ 7.82−7.77 (*m*, 2H), 7.55−7.51 (*m*, 2H), 6.91 (*d*, J = 14.4 Hz, 1H), 4.87 (*dt*, J = 14.4, 7.7 Hz, 1H), 3.81 (*dd*, J = 7.7, 1.1 Hz, 2H), 3.56−3.47 (*m*, 2H), 2.56−2.44 (*m*, 2H), 2.19−2.06 (*m*, 2H). ^{13}C{^{1}H} NMR (101 MHz, $CDCl_3$): δ 173.4, 140.7, 136.9, 131.3, 129.9, 129.6, 96.0, 58.6, 45.0, 30.9, 17.4. HRMS (ESI) m/z: $[M + H]^+$ Calcd for $C_{13}H_{15}ClNO_3S$ 300.0456, found 300.0459. $[M + Na]^+$ Calcd for $C_{13}H_{14}ClNO_3SNa$ 322.0269, found 322.0275. m.p.: 205−208 °C

3ae *(E)-1-(3-((3-Chloro-4-methylphenyl)sulfonyl)prop-1-en-1-yl)pyrrolidin-2-one* $C_{14}H_{16}ClNO_3S$. The reaction of *N*-allenyl-2-pyrrolidinone (**1a**, 0.25 mmol) with 3-chloro-4-methylbenzene sulfinic acid sodium salt (**2e**, 2 equiv, 0.5 mmol) afforded the corresponding hydrosulfonylation product **3ae** as a white solid (62.8 mg, 80%) according to the general procedure. ^{1}H NMR (400 MHz, $CDCl_3$): δ 7.86−7.80 (*m*, 1H), 7.68−7.60 (*m*, 1H), 7.45−7.38 (*m*, 1H), 6.87 (*d*, J = 14.4 Hz, 1H), 4.87 (*dt*, J = 14.4, 7.7 Hz, 1H), 3.79 (*dd*, J = 7.7, 1.0 Hz, 2H), 3.52 (*m*, 2H), 2.51−2.47 (*m*, 2H), 2.46 (*s*, 3H), 2.18−2.09 (*m*, 2H). ^{13}C{^{1}H} NMR (101 MHz, $CDCl_3$): δ 173.4, 143.0, 137.1, 135.4, 131.6, 131.3, 129.1, 126.4, 96.2, 58.7, 45.1, 30.9, 20.5, 17.4. HRMS (ESI) m/z: $[M + H]^+$ Calcd for $C_{14}H_{17}ClNO_3S$ 314.0612, found 314.0614. $[M + Na]^+$ Calcd for $C_{14}H_{16}ClNO_3SNa$ 336.0432, found 336.0425. m.p.: 177−181 °C

3af *(E)-1-(3-((3-(Trifluoromethyl)phenyl)sulfonyl)prop-1-en-1-yl)pyrrolidin-2-one* $C_{14}H_{14}F_3NO_3S$. The reaction of *N*-allenyl-2-pyrrolidinone (**1a**, 0.25 mmol) with 3-trifluoromethylbenzene sulfinic acid sodium salt (**2f**, 2 equiv, 0.5 mmol) afforded the corresponding hydrosulfonylation product **3af** as a white solid (74.2 mg, 89%) according to the general procedure. ^{1}H{^{19}F} NMR (400 MHz, $CDCl_3$): δ 8.11−8.05 (*m*, 2H), 7.94−7.89 (*m*, 1H), 7.73 (*t*, J = 7.6 Hz, 2H), 6.81 (*d*, J = 14.4 Hz, 1H), 4.87 (*dt*, J = 14.5, 7.8 Hz, 1H), 3.83 (*dd*, J = 7.7, 1.0 Hz, 2H), 3.56−3.46 (*m*, 2H), 2.46 (*t*, J = 8.2 Hz, 2H), 2.18−2.08 (*m*, 2H). ^{13}C{^{1}H} NMR (101 MHz, $CDCl_3$): δ 173.4 (*s*), 139.3 (*s*), 131.7 (*s*), 131.6 (*q*, J = 33.6 Hz), 131.6 (*q*, J = 1.1 Hz), 130.5 (*q*, J = 3.6 Hz), 130.1 (*s*), 126.0 (*q*, J = 3.8 Hz), 126.0 (*q*, J = 272 Hz), 95.7 (*s*), 58.6 (*s*), 44.9 (*s*), 30.8 (*s*), 17.4 (*s*). ^{19}F{^{1}H} NMR (377 MHz, $CDCl_3$): δ −62.87 (*s*). HRMS (ESI) m/z: $[M + H]^+$ Calcd for $C_{14}H_{15}F_3NO_3S$ 334.0719, found 334.0723. $[M + Na]^+$ Calcd for $C_{14}H_{14}F_3NO_3SNa$ 356.0539, found 356.0532. m.p.: 142−145 °C

https://doi.org/10.1021/acs.joc.1c01345
J. Org. Chem. 2021, 86, 15695−15701

3ag *(E)-1-(3-(Pyridin-2-ylsulfonyl)prop-1-en-1-yl)pyrrolidin-2-one* $C_{12}H_{14}N_2O_3S$. The reaction of *N*-allenyl-2-pyrrolidinone (**1a**, 0.25 mmol) with pyridine-2-sulfinic acid sodium salt (**2g**, 2 equiv, 0.5 mmol) afforded the corresponding hydrosulfonylation product **3ag** as a pale brown oil (39.9 mg, 60%) according to a modified procedure using 2 equiv of TFA instead of 1 equiv. ^{1}H NMR (400 MHz, $CDCl_3$): δ 8.77 (*m*, 1H), 8.11−8.02 (*m*, 1H), 7.94 (*m*, 1H), 7.55 (*m*, 1H), 6.95 (*d*, *J* = 14.4 Hz, 1H), 4.89 (*dt*, *J* = 14.4, 7.8 Hz, 1H), 4.16 (*dd*, *J* = 7.8, 1.1 Hz, 2H), 3.54−3.43 (*m*, 2H), 2.44 (*m*, 2H), 2.16−2.01 (*m*, 2H). ^{13}C{^{1}H} NMR (101 MHz, $CDCl_3$): δ 173.3, 156.8, 150.3, 138.0, 131.4, 127.4, 122.6, 95.7, 54.1, 44.9, 30.9, 17.4. HRMS (ESI) *m*/*z*: [M + H]$^+$ Calcd for $C_{12}H_{15}N_2O_3S$ 267.0798, found 267.0805. [M + Na]$^+$ Calcd for $C_{12}H_{14}N_2O_3SNa$ 289.0617, found 289.0619. m.p.: 138−140 °C

3ah *(E)-1-(3-((2,5-Dichlorothiophen-3-yl)sulfonyl)prop-1-en-1-yl)pyrrolidin-2-one* $C_{11}H_{11}Cl_2NO_3S_2$. The reaction of *N*-allenyl-2-pyrrolidinone (**1a**, 0.25 mmol) with 2,5-dichlorothiophene-3-sulfinic acid sodium salt (**2h**, 2 equiv, 0.5 mmol) afforded the corresponding hydrosulfonylation product **3ah** as a white solid (70.6 mg, 83%) according to the general procedure. ^{1}H NMR (400 MHz, $CDCl_3$): δ 7.10 (*s*, 1H), 6.99 (*d*, *J* = 14.4 Hz, 1H), 4.89 (*dt*, *J* = 14.4, 7.8 Hz, 1H), 3.93 (*dd*, *J* = 7.8, 1.1 Hz, 2H), 3.57−3.47 (*m*, 2H), 2.55−2.45 (*m*, 2H), 2.19−2.07 (*m*, 2H). ^{13}C{^{1}H} NMR (101 MHz, $CDCl_3$): δ 173.4, 134.8, 132.7, 131.6, 127.9, 126.8, 95.3, 58.1, 44.7, 30.9, 17.4. HRMS (ESI) *m*/*z*: [M + H]$^+$ Calcd for $C_{11}H_{12}Cl_2NO_3S_2$339.9630, found 339.9634. [M + Na]$^+$ Calcd for $C_{11}H_{11}Cl_2NO_3S_2Na$361.9450, found 361.9446. m.p.: 158−161 °C

3ai *(E)-1-(3-(Methylsulfonyl)prop-1-en-1-yl)pyrrolidin-2-one* $C_8H_{13}NO_3S$. The reaction of *N*-allenyl-2-pyrrolidinone (**1a**, 0.25 mmol) with methanesulfinic acid sodium salt (**2i**, 2 equiv, 0.5 mmol) afforded the corresponding hydrosulfonylation product **3ai** as a pale yellow waxy solid (32.0 mg, 63%) according to the general procedure. ^{1}H NMR (400 MHz, $CDCl_3$): δ 7.16 (*d*, *J* = 14.4 Hz, 1H), 4.99 (*dt*, *J* = 14.4, 7.8 Hz, 1H), 3.73 (*dd*, *J* = 7.8, 1.1 Hz, 2H), 3.61−3.51 (*m*, 2H), 2.92−2.77 (*s*, 3H), 2.57−2.46 (*m*, 2H), 2.20−2.09 (*m*, 2H). ^{13}C{^{1}H} NMR (101 MHz, $CDCl_3$): δ 173.6, 130.9, 97.1, 57.0, 45.0, 38.9, 31.0, 17.4. HRMS (ESI) *m*/*z*: [M + H]$^+$ Calcd for $C_8H_{14}NO_3S$ 204.0694, found 204.0684.

3aj *(E)-1-(3-(Isopropylsulfonyl)prop-1-en-1-yl)pyrrolidin-2-one* $C_{10}H_{17}NO_3S$. The reaction of *N*-allenyl-2-pyrrolidinone (**1a**, 0.25 mmol) with isopropylsulfinic acid sodium salt (**2j**, 2 equiv, 0.5 mmol) afforded the corresponding hydrosulfonylation product **3aj** as a brown waxy solid (35.3 mg, 61%) according to the general procedure. ^{1}H NMR (400 MHz, $CDCl_3$): δ 7.12 (*d*, *J* = 14.4 Hz, 1H), 4.95 (*dt*, *J* = 14.5, 7.7 Hz, 1H), 3.70 (*dd*, *J* = 7.7, 1.1 Hz, 2H), 3.58−3.52 (*m*, 2H), 3.18 (*hept*, *J* = 6.9 Hz, 1H), 2.52−2.46 (*m*, 2H), 2.17−2.07 (*m*, 2H), 1.36 (*d*, *J* = 6.9 Hz, 6H). ^{13}C{^{1}H} NMR (101 MHz, $CDCl_3$): δ 173.6, 130.5, 96.7, 51.9, 51.1, 45.1, 31.0, 17.4, 15.2. HRMS (ESI) *m*/*z*: [M + H]$^+$ Calcd for $C_{10}H_{18}NO_3S$ 232.1002, found 232.1011. [M + Na]$^+$ Calcd for $C_{10}H_{17}NO_3SNa$ 254.0821, found 254.0824.

3ak *(E)-1-(3-(Cyclopropylsulfonyl)prop-1-en-1-yl)pyrrolidin-2-one* $C_{10}H_{15}NO_3S$. The reaction of *N*-allenyl-2-pyrrolidinone (**1a**, 0.25 mmol) with cyclopropylsulfinic acid sodium salt (**2k**, 2 equiv, 0.5 mmol) afforded the corresponding hydrosulfonylation product **3ak** as a pale yellow solid (49.3 mg, 90%) according to the general procedure. ^{1}H NMR (400 MHz, $CDCl_3$): δ 7.17 (*d*, *J* = 14.4 Hz, 1H), 5.01 (*dt*, *J* = 14.4, 7.8 Hz, 1H), 3.76 (*dd*, *J* = 7.8, 1.1 Hz, 2H), 3.60−3.54 (*m*, 2H), 2.55−2.48 (*m*, 2H), 2.40 (*tt*, *J* = 8.0, 4.9 Hz, 1H), 2.20−2.10 (*m*, 2H), 1.27−1.20 (*m*, 2H), 1.08−1.00 (*m*, 2H). ^{13}C{^{1}H} NMR (101 MHz, $CDCl_3$): δ 173.5, 130.7, 97.0, 56.01, 45.1, 31.0, 28.2, 17.4, 4.8. HRMS (ESI) *m*/*z*: [M + H]$^+$ Calcd for $C_{10}H_{16}NO_3S$ 230.0845, found 230.0851. [M + Na]$^+$ Calcd for $C_{10}H_{15}NO_3SNa$ 252.0665, found 252.0663. m.p.: 109−113 °C

3al *(E)-1-(3-(Cyclohexylsulfonyl)prop-1-en-1-yl)pyrrolidin-2-one* $C_{13}H_{21}NO_3S$. The reaction of *N*-allenyl-2-pyrrolidinone (**1a**, 0.25 mmol) with cyclohexylsulfinic acid sodium salt (**2l**, 2 equiv, 0.5 mmol) afforded the corresponding hydrosulfonylation product **3al** as a white solid (54.3 mg, 80%) after silica gel column chromatography (AcOEt/*n*-hex, 20:80) according to the general procedure. ^{1}H NMR (400 MHz, $CDCl_3$): δ 7.08 (*d*, *J* = 14.4 Hz, 1H), 4.91 (*dt*, *J* = 15.2, 7.7 Hz, 1H), 3.65 (*d*, *J* = 7.7 Hz, 2H), 3.53 (*t*, *J* = 7.2 Hz, 2H), 2.88 (*tt*, *J* = 12.0, 3.3 Hz, 1H), 2.46 (*t*, *J* = 8.2 Hz, 2H), 2.15−2.09 (*m*, 2H), 2.07 (*m*, 2H), 1.88 (*m*, 2H), 1.68 (*m*, 1H), 1.58−1.44 (*m*, 2H), 1.32−1.11 (*m*, 3H). ^{13}C{^{1}H} NMR (101 MHz, $CDCl_3$): δ 173.5, 130.5, 96.6, 59.2, 51.9, 45.0, 31.0, 25.0, 25.0, 24.9, 17.4. HRMS (ESI) *m*/*z*: [M + H]$^+$ Calcd for $C_{13}H_{22}NO_3S$ 272.1315, found 272.1321. [M + Na]$^+$ Calcd for $C_{13}H_{21}NO_3SNa$ 294.1134, found 294.1134. m.p.: 141−144 °C

3am *(E)-1-(3-((Trifluoromethyl)sulfonyl)prop-1-en-1-yl)-pyrrolidin-2-one* $C_8H_{10}F_3NO_3S$. The reaction of *N*-allenyl-2-pyrrolidinone (**1a**, 0.25 mmol) with sodium triflinate (**2m**, 2 equiv, 0.5 mmol) afforded the corresponding hydrosulfonylation product **3am** as a white solid (46.9 mg, 73%) according to the general procedure. ^{1}H{^{19}F} NMR (400 MHz, $CDCl_3$): δ 7.24 (*d*, *J* = 14.6 Hz, 1H), 4.81 (*dt*, *J* = 14.6, 7.8 Hz, 1H), 4.01 (*dd*, *J* = 7.7, 1.0 Hz, 2H), 3.60−3.50 (*m*, 2H), 2.51 (*m*, 2H), 2.21−2.10 (*m*, 2H). ^{13}C{^{1}H} NMR (101 MHz, $CDCl_3$): δ 173.7 (*s*), 133.1 (*s*), 119.7 (*q*, *J* = 328.7 Hz), 91.2 (*s*), 52.7 (*s*), 44.9 (*s*), 30.8 (*s*), 17.4 (*s*). ^{19}F{^{1}H} NMR (377 MHz, $CDCl_3$): δ −76.09 (*s*). HRMS (ESI) *m*/*z*: [M + H]$^+$ Calcd for $C_8H_{11}F_3NO_3S$ 258.0406, found 258.0411. [M + Na]$^+$ Calcd for $C_8H_{10}F_3NO_3SNa$ 280.0226, found 280.0216. m.p.: 88−92 °C

3an *Methyl (E)-3-((3-(2-Oxopyrrolidin-1-yl)allyl)sulfonyl)-propanoate* $C_{11}H_{17}NO_5S$. The reaction of *N*-allenyl-2-pyrrolidinone (**1a**, 0.25 mmol) with sodium 1-methyl 3-sulfinopropanoate (**2n**, 2 equiv, 0.5 mmol) afforded the corresponding hydrosulfonylation product **3an** as a white solid (48.2 mg, 70%) according to the general procedure. ^{1}H NMR (400 MHz, $CDCl_3$): δ 7.18 (*d*, *J* = 14.4 Hz, 1H), 4.95 (*dt*, *J* = 14.8, 7.7 Hz, 1H), 3.76 (*d*, *J* = 7.7 Hz, 2H), 3.72 (*s*, 3H), 3.55 (*t*, *J* = 7.2 Hz, 2H), 3.28 (*t*, *J* = 7.4 Hz, 2H), 2.84 (*t*, *J* = 7.4 Hz, 2H), 2.50 (*t*, *J* = 8.1 Hz, 2H), 2.19−2.08 (*m*, 2H). ^{13}C{^{1}H} NMR (101 MHz, $CDCl_3$): δ 173.6, 170.9, 131.1, 96.4, 55.8, 52.5, 46.3, 45.1, 31.0, 26.6, 17.4. HRMS (ESI) *m*/*z*: [M + H]$^+$ Calcd for $C_{11}H_{18}NO_5S$ 276.0900, found 276.0905. [M + Na]$^+$ Calcd for $C_{11}H_{17}NO_5SNa$ 298.0733, found 298.0720. m.p.: 104−107 °C

3ba *(E)-3-(3-(Phenylsulfonyl)prop-1-en-1-yl)oxazolidin-2-one* $C_{12}H_{13}NO_4S$. The reaction of *N*-allenyl-2-oxazolidinone (**1b**, 0.25 mmol) with benzenesulfinic acid sodium salt (**2a**, 2 equiv, 0.5 mmol) afforded the corresponding hydrosulfonylation product **3ba** as a white solid (60.1 mg, 86%) according to the general procedure. ^{1}H NMR (400 MHz, $CDCl_3$): δ 7.90−7.85 (*m*, 2H), 7.70−7.64 (*m*, 1H), 7.60−7.54 (*m*, 2H), 6.68 (*d*, *J* = 14.3 Hz, 1H), 4.81 (*dt*, *J* = 14.2, 7.8 Hz, 1H), 4.50−4.43 (*m*, 2H), 3.80 (*dd*, *J* = 7.8, 1.1 Hz, 2H), 3.75−3.68 (*m*, 2H). ^{13}C{^{1}H} NMR (101 MHz, $CDCl_3$): δ 154.9, 138.3, 134.0, 131.5, 129.3, 128.3, 95.6, 62.3, 58.1, 42.3. HRMS (ESI) *m*/*z*: [M + H]$^+$ Calcd for $C_{12}H_{14}NO_4S$ 268.0638, found 268.0643. [M + Na]$^+$ Calcd for $C_{12}H_{13}NO_4SNa$ 290.0457, found 290.0452. m.p.: 135−138 °C

3bb *(E)-3-(3-Tosylprop-1-en-1-yl)oxazolidin-2-one* $C_{13}H_{15}NO_4S$. The reaction of *N*-allenyl-2-oxazolidinone (**1b**, 0.25 mmol) with *p*-toluenesulfinic acid sodium salt (**2b**, 2 equiv, 0.5 mmol) afforded the corresponding hydrosulfonylation product **3bb** as a white solid (56.3 mg, 80%) according to the general procedure. ^{1}H NMR (400 MHz, $CDCl_3$): δ 7.76−7.71 (*m*, 2H), 7.37−7.31 (*m*, 2H), 6.66 (*d*, *J* = 14.3 Hz, 1H), 4.80 (*dt*, *J* = 14.3, 7.7 Hz, 1H), 4.49−4.42 (*m*, 2H), 3.77 (*dd*, *J* = 7.7, 0.9 Hz, 2H), 3.71 (*dd*, *J* = 8.8, 7.3 Hz, 2H), 2.45 (*s*, 3H). ^{13}C{^{1}H} NMR (101 MHz, $CDCl_3$): δ 155.0, 145.1, 135.4, 131.4, 130.0, 128.4, 96.1, 62.4, 58.3, 42.4, 21.8. HRMS (ESI) *m*/*z*: [M + H]$^+$ Calcd for $C_{13}H_{16}NO_4S$ 282.0800, found 282.0802. [M + Na]$^+$ Calcd for $C_{13}H_{15}NO_4SNa$ 304.0619, found 304.0632. m.p.: 173−177 °C

3bi *(E)-3-(3-(Isopropylsulfonyl)prop-1-en-1-yl)oxazolidin-2-one* $C_7H_{11}NO_4S$. The reaction of *N*-allenyl-2-oxazolidinone (**1b**, 0.25 mmol) with methanesulfinic acid sodium salt (**2i**, 2 equiv, 0.5 mmol) afforded the corresponding hydrosulfonylation product **3bi** as a white solid (26.2 mg, 51%) according to the general procedure. ^{1}H NMR (400 MHz, $CDCl_3$): δ 6.99 (*d*, *J* = 14.3 Hz, 1H), 4.91 (*dt*, *J* = 14.3, 7.8 Hz, 1H), 4.53−4.48 (*m*, 2H), 3.78 (*dd*, *J* = 8.9, 7.2 Hz, 2H), 3.74 (*d*, *J* = 7.8 Hz, 2H), 2.88 (*s*, 3H). ^{13}C{^{1}H} NMR (101 MHz, $CDCl_3$): δ 155.2, 131.5, 96.3, 62.5, 56.6, 42.4, 39.1. HRMS (ESI) *m*/*z*: [M + Na]$^+$ Calcd for $C_7H_{11}NO_4Na$ 228.0306, found 228.0313. m.p.: 156−158 °C

https://doi.org/10.1021/acs.joc.1c01345
J. Org. Chem. 2021, 86, 15695−15701

3bj *(E)-3-(3-(Isopropylsulfonyl)prop-1-en-1-yl)oxazolidin-2-one* $C_9H_{15}NO_4S$. The reaction of *N*-allenyl-2-oxazolidinone (**1b**, 0.25 mmol) with isopropylsulfinic acid sodium salt (**2j**, 2 equiv, 0.5 mmol) afforded the corresponding hydrosulfonylation product **3bj** as a pale yellow solid (33.2 mg, 57%) according to the general procedure. ^{1}H NMR (400 MHz, $CDCl_3$): δ 6.96 (*d*, *J* = 14.3 Hz, 1H), 4.88 (*dt*, *J* = 14.4, 7.7 Hz, 1H), 4.52−4.46 (*m*, 2H), 3.78 (*dd*, *J* = 8.9, 7.2 Hz, 2H), 3.71 (*dd*, *J* = 7.7, 1.1 Hz, 2H), 3.20 (*hept*, *J* = 6.9 Hz, 1H), 1.40 (*d*, *J* = 6.9 Hz, 6H). ^{13}C{^{1}H} NMR (101 MHz, $CDCl_3$): δ 155.0, 131.0, 95.7, 62.4, 51.4, 51.4, 42.3, 15.3. HRMS (ESI) *m*/*z*: $[M + Na]^+$ Calcd for $C_9H_{15}NO_4SNa$ 256.0619, found 256.0630. m.p.: 190−192 °C

3bm *(E)-3-(3-((Trifluoromethyl)sulfonyl)prop-1-en-1-yl)-oxazolidin-2-one* $C_7H_8F_3NO_4S$. The reaction of *N*-allenyl-2-oxazolidinone (**1b**, 0.25 mmol) with sodium triflinate (**2m**, 2 equiv, 0.5 mmol) afforded the corresponding hydrosulfonylation product **3bm** as a white solid (40.8 mg, 63%) according to the general procedure. ^{1}H{^{19}F} NMR (400 MHz, $CDCl_3$): δ 7.08 (*d*, *J* = 14.2 Hz, 1H), 4.74 (*dt*, *J* = 14.2, 7.8 Hz, 1H), 4.55−4.48 (*m*, 2H), 4.01 (*dd*, *J* = 7.8, 1.1 Hz, 2H), 3.82−3.75 (*m*, *J* = 8.9, 7.2 Hz, 2H). ^{13}C{^{1}H} NMR (101 MHz, $CDCl_3$): δ 154.8 (*s*), 133.7 (*s*), 119.7 (*q*, *J* = 328.4 Hz), 90.6 (*s*), 62.4 (*s*), 52.3 (*s*), 42.2 (*s*). ^{19}F{^{1}H} NMR (377 MHz, $CDCl_3$): δ −76.02 (*s*). HRMS (ESI) *m*/*z*: $[M + H]^+$ Calcd for $C_7H_9F_3NO_4S$ 260.0199, found 260.0210. $[M + Na]^+$ Calcd for $C_7H_8F_3NO_4SNa$ 282.0018, found 282.0027. m.p.: 118−120 °C

3ca *1,3-Bis((E)-3-(phenylsulfonyl)prop-1-en-1-yl)imidazolidin-2-one* $C_{21}H_{22}N_2O_5S_2$. The reaction of *N*,*N*-diallenyl-2-imidazolidinone (**1c**, 0.25 mmol) with benzenesulfinic acid sodium salt (**2a**, 2 equiv, 0.5 mmol) afforded the corresponding hydrosulfonylation product **3ca** as a white solid (70 mg, 60%) according to a modified procedure using 2 equiv of TFA and 4 equiv of $PhSO_2Na$ in a H_2O/THF 1:1 (0.5 M) mixture as the solvent. ^{1}H NMR (400 MHz, $CDCl_3$): δ 7.89−7.83 (*m*, 4H), 7.69−7.62 (*m*, 2H), 7.59−7.52 (*m*, 4H), 6.76 (*d*, *J* = 14.3 Hz, 2H), 4.70 (*dt*, *J* = 14.3, 7.7 Hz, 2H), 3.79 (*dd*, *J* = 7.7, 0.9 Hz, 4H), 3.62 (s, 4H). ^{13}C{^{1}H} NMR (101 MHz, $CDCl_3$): δ 152.3, 138.5, 134.0, 131.7, 129.3, 128.4, 93.4, 58.5, 39.8. HRMS (ESI) *m*/*z*: $[M + H]^+$ Calcd for $C_{21}H_{23}N_2O_5S_2$ 447.1043, found 447.1048. m.p.: 182−183 °C

3da *(E)-4-Methyl-N-phenyl-N-(3-(phenylsulfonyl)prop-1-en-1-yl)benzenesulfonamide* $C_{22}H_{21}NO_4S_2$. The reaction of *N*-allenyl-*N*-tosylaniline (**1d**, 0.25 mmol) with benzenesulfinic acid sodium salt (**2a**, 2 equiv, 0.5 mmol) afforded the corresponding hydrosulfonylation product **3da** as a pale brown oil (64.1 mg, 60%) according to the general procedure. ^{1}H NMR (400 MHz, $CDCl_3$): δ 7.79−7.74 (*m*, 2H), 7.69−7.64 (*m*, 1H), 7.57−7.50 (*m*, 2H), 7.48−7.43 (*m*, 2H), 7.42−7.33 (*m*, 3H), 7.31−7.26 (*m*, 2H), 7.05 (*d*, *J* = 13.9 Hz, 1H), 6.91−6.86 (*m*, 2H), 4.22 (*dt*, *J* = 13.9, 7.9 Hz, 1H), 3.72 (*dd*, *J* = 7.9, 0.9 Hz, 2H), 2.47 (*s*, *J* = 15.8 Hz, 3H). ^{13}C{^{1}H} NMR (101 MHz, $CDCl_3$): δ 144.4, 138.1, 136.7, 135.5, 135.3, 133.7, 130.0, 129.8, 129.7, 129.5, 129.1, 128.5, 127.5, 96.1, 58.1, 21.7. HRMS (ESI) *m*/*z*: $[M + H]^+$ Calcd for $C_{22}H_{22}NO_4S_2$428.0985, found 428.0982. $[M + Na]^+$ Calcd for $C_{22}H_{21}NO_4S_2Na$450.0804, found 450.0791.

3db *(E)-4-Methyl-N-phenyl-N-(3-tosylprop-1-en-1-yl)-benzenesulfonamide* $C_{23}H_{23}NO_4S_2$. The reaction of *N*-allenyl-*N*-tosylaniline (**1d**, 0.25 mmol) with *p*-toluenesulfinic acid sodium salt (**2b**, 2 equiv, 0.5 mmol) afforded the corresponding hydrosulfonylation product **3db** as a pale brown oil (38.6 mg, 35%) according to the general procedure. ^{1}H NMR (400 MHz, $CDCl_3$): δ 7.65−7.59 (*m*, 2H), 7.49−7.42 (*m*, 2H), 7.40−7.32 (*m*, 3H), 7.32−7.28 (*m*, 2H), 7.28−7.24 (*m*, 2H), 7.02 (*d*, *J* = 13.9 Hz, 1H), 6.89−6.85 (*m*, 2H), 4.21 (*dt*, *J* = 13.9, 7.9 Hz, 1H), 3.66 (*dd*, *J* = 7.9, 0.9 Hz, 2H), 2.46 (*s*, 3H), 2.45 (*s*, 3H). ^{13}C{^{1}H} NMR (101 MHz, $CDCl_3$): δ 144.8, 144.4, 136.7, 135.7, 135.4, 135.4, 130.1, 129.8, 129.8, 129.6, 128.6, 127.7, 121.8, 96.4, 58.3, 21.8, 21.8. HRMS (ESI) *m*/*z*: $[M + H]^+$ Calcd for $C_{23}H_{24}NO_4S_2$442.1141, found 442.1135.

4aa. *(R,S)-1-(1,3-Bis(phenylsulfonyl)propyl)pyrrolidin-2-one* $C_{19}H_{21}NO_5S_2$. The reaction of *N*-allenyl-2-pyrrolidinone (**1a**, 0.25 mmol) with benzenesulfinic acid sodium salt (**2a**, 4 equiv, 1 mmol) afforded the corresponding double hydrosulfonylation product **4aa** as an off-white solid (83.54 mg, 82%) according to a modified procedure. The reaction was performed at 40 °C over wto days under open air with 4 equiv of TFA, 4 equiv of $PhSO_2Na$, and H_2O/THF 1:1 (0.25M) as the solvent. ^{1}H NMR (400 MHz, $CDCl_3$): δ 7.93−7.89 (*m*, 2H), 7.85−7.81 (*m*, 2H), 7.73−7.64 (*m*, 2H), 7.63−7.52 (*m*, 4H), 5.24 (*dd*, *J* = 10.5, 4.7 Hz, 1H), 3.79−3.72 (*m*, 1H), 3.39−3.32 (*m*, 1H), 3.22 (*ddd*, *J* = 14.1, 10.9, 6.2 Hz, 1H), 2.94 (*ddd*, *J* = 14.2, 10.7, 4.4 Hz, 1H), 2.61−2.41 (app. *m*, expected 2x *dddd*, 2H), 2.27−2.16 (*m*, 1H), 2.05−1.90 (*m*, 3H). ^{13}C{^{1}H} NMR (101 MHz, $CDCl_3$): δ 175.5, 138.3, 136.2, 134.8, 134.4, 129.8, 129.4, 128.8, 128.2, 69.4, 52.4, 42.9, 31.1, 30.1, 18.5, 18.3. HRMS (ESI) *m*/*z*: $[M + H]^+$ Calcd for $C_{19}H_{22}NO_5S_2$408.0939, found 408.0945. m.p.: 197−199 °C

Intermediate A. (*E*)-3-(2-Oxopyrrolidin-1-yl)allyl 2,2,2-trifluoroacetate $C_9H_{10}F_3NO_3$. MW: 237.18 g/mol. ^{1}H NMR (400 MHz, *d*-THF): δ 7.32 (*d*, *J* = 14.1 Hz, 1H), 5.13 (*dt*, 14.1, 8.1 Hz, 1H), 4.95 (*dd*, *J* = 7.8 Hz, 2H), 3.57−3.50 (*m*, 2H), 2.41 (*m*, 2H), 2.11 (*m*, 2H). ^{19}F{^{1}H} NMR (377 MHz, *d*-THF): δ −76.31 (*s*).

Intermediate B. (*E*)-1-Allylidene-2-oxopyrrolidin-1-ium $C_7H_{10}NO^+$. MW: 124.16 g/mol. HRMS (ESI) *m*/*z*: $[M]^+$ Calcd for $C_7H_{10}NO$ 124.0757, found 124.0758.

d-3aa *(E)-1-(3-(Phenylsulfonyl)prop-1-en-1-yl-2-d)pyrrolidin-2-one* $C_{13}H_{14}DNO_3S$. The reaction of *N*-allenyl-2-pyrrolidinone (**1a**, 0.25 mmol) with benzenesulfinic acid sodium salt (**2a**, 2 equiv, 0.5 mmol) in D_2O afforded a 90:10 mixture of ***d*-3aa** and **3aa**. ^{1}H NMR (400 MHz, $CDCl_3$): δ 7.87−7.77 (*m*, 2H), 7.66−7.58 (*m*, 1H), 7.51 (*m*, 2H), 6.83 (*s*, 1H), 3.77 (*s*, 2H), 3.45 (*t*, *J* = 7.2 Hz, 1H), 2.41 (*t*, *J* = 8.1 Hz, 1H), 2.14−2.00 (*m*, 1H).

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.joc.1c01345.

^{1}H and ^{13}C NMR spectra (PDF)

FAIR data, including the primary NMR FID files, for compounds **3aa**−**3an**, **3ba**, **3bb**, **3bi**, **3bj**, **3bm**, **3ca**, **3da**, **3db**, **4aa**, and ***d*-3aa** (ZIP)

■ AUTHOR INFORMATION

Corresponding Authors

Florian Monnier − *Institut Charles Gerhardt Montpellier, University of Montpellier, CNRS, ENSCM, Montpellier 34296, France; Institut Universitaire de France (IUF), Paris 75231 CEDEX 5, France*; Email: florian.monnier@enscm.fr

Marc Taillefer − *Institut Charles Gerhardt Montpellier, University of Montpellier, CNRS, ENSCM, Montpellier 34296, France*; Email: marc.taillefer@enscm.fr

Authors

Lucas Pagès − *Institut Charles Gerhardt Montpellier, University of Montpellier, CNRS, ENSCM, Montpellier 34296, France*

Sébastien Lemouzy − *Institut Charles Gerhardt Montpellier, University of Montpellier, CNRS, ENSCM, Montpellier 34296, France*

Complete contact information is available at: https://pubs.acs.org/10.1021/acs.joc.1c01345

Notes

The authors declare no competing financial interest.

■ ACKNOWLEDGMENTS

This publication is dedicated to Dr. Christian Bruneau as a tribute to his contribution to the chemistry of catalysis for a sustainable development. We thank Gilles Valette for technical

assistance on HRMS. Financial support was provided by ENSCM (Ph.D. fellowship for L.P. and postdoctoral fellowship for S.L.) and CNRS. IUF is warmly acknowledged with thanks.

REFERENCES

(1) Chen, X.; Hussain, S.; Parveen, S.; Zhang, S.; Yang, Y.; Zhu, C. Sulfonyl Group-Containing Compounds in the Design of Potential Drugs for the Treatment of Diabetes and Its Complications. *Curr. Med. Chem.* **2012**, *19* (21), 3578−3604.
(2) Scott, K. A.; Njardarson, J. T. Analysis of US FDA-Approved Drugs Containing Sulfur Atoms. *Top. Curr. Chem.* **2018**, *376* (1), 5.
(3) El-Awa, A.; Noshi, M. N.; du Jourdin, X. M.; Fuchs, P. L. Evolving Organic Synthesis Fostered by the Pluripotent Phenylsulfone Moiety. *Chem. Rev.* **2009**, *109* (12), 6920−6921.
(4) Alba, A. N. R.; Companyó, X.; Rios, R. Sulfones: New Reagents in Organocatalysis. *Chem. Soc. Rev.* **2010**, *39* (6), 2018−2033.
(5) Back, T. G.; Clary, K. N.; Gao, D. Cycloadditions and Cyclizations of Acetylenic, Allenic, and Conjugated Dienyl Sulfones. *Chem. Rev.* **2010**, *110* (8), 4498−4553.
(6) Buynak, J. D.; Doppalapudi, V. R.; Rao, A. S.; Nidamarthy, S. D.; Adam, G. The Synthesis and Evaluation of 2-Substituted-7-(Alkylidene)Cephalosporin Sulfones as β-Lactamase Inhibitors. *Bioorg. Med. Chem. Lett.* **2000**, *10* (9), 847−851.
(7) Reck, F.; Zhou, F.; Girardot, M.; Kern, G.; Eyermann, C. J.; Hales, N. J.; Ramsay, R. R.; Gravestock, M. B. Identification of 4-Substituted 1,2,3-Triazoles as Novel Oxazolidinone Antibacterial Agents with Reduced Activity against Monoamine Oxidase A. *J. Med. Chem.* **2005**, *48* (2), 499−506.
(8) Nielsen, M.; Jacobsen, C. B.; Holub, N.; Paixão, M. W.; Jørgensen, K. A. Asymmetric Organocatalysis with Sulfones. *Angew. Chem., Int. Ed.* **2010**, *49* (15), 2668−2679.
(9) Trost, B. M.; Kalnmals, C. A. Sulfones as Chemical Chameleons: Versatile Synthetic Equivalents of Small-Molecule Synthons. *Chem. - Eur. J.* **2019**, *25* (48), 11193−11213.
(10) Clayden, J.; Julia, M. Allylic Sulfones as Allyl Anion Equivalents: Homoallylic Alcohols from Metal Catalysed Reactions of Sulfones with Aldehydes and Ketones. *J. Chem. Soc., Chem. Commun.* **1994**, No. 16, 1905−1906.
(11) Blieck, R.; Taillefer, M.; Monnier, F. Metal-Catalyzed Intermolecular Hydrofunctionalization of Allenes: Easy Access to Allylic Structures via the Selective Formation of C-N, C-C, and C-O Bonds. *Chem. Rev.* **2020**, *120* (24), 13545−13598.
(12) Kamijo, S.; Al-Masum, M.; Yamamoto, Y. Palladium Catalyzed Hydrosulfination of Allenes with Tosylhydrazine Leading to Allylsulfones. *Tetrahedron Lett.* **1998**, *39* (7), 691−694.
(13) Khakyzadeh, V.; Wang, Y. H.; Breit, B. Rhodium-Catalyzed Addition of Sulfonyl Hydrazides to Allenes: Regioselective Synthesis of Branched Allylic Sulfones. *Chem. Commun.* **2017**, *53* (36), 4966−4968.
(14) Pritzius, A. B.; Breit, B. Asymmetric Rhodium-Catalyzed Addition of Thiols to Allenes: Synthesis of Branched Allylic Thioethers and Sulfones. *Angew. Chem., Int. Ed.* **2015**, *54* (10), 3121−3125.
(15) Miao, M.; Luo, Y.; Xu, H.; Chen, Z.; Xu, J.; Ren, H. Solvent-Controlled, Tunable Hydrosulfonylation of 3-Cyclopropylideneprop-2-En-1-Ones. *Org. Lett.* **2016**, *18* (17), 4292−4295.
(16) Miao, M.; Xu, H.; Luo, Y.; Jin, M.; Chen, Z.; Xu, J.; Ren, H. A Modular Approach to Highly Functionalized 3-Sulfonylfurans: Via Conjugate Addition of 3-Cyclopropylideneprop-2-En-1-Ones with Sodium Sulfinates and Sequential 5- Endo -Trig Iodocyclization. *Org. Chem. Front.* **2017**, *4* (9), 1824−1828.
(17) Goh, J.; Maraswami, M.; Loh, T.-P. Synthesis of Vinylic Sulfones in Aqueous Media. *Org. Lett.* **2021**, *23*, 1060−1065.
(18) Liu, Y.; Xie, P.; Sun, Z.; Wo, X.; Gao, C.; Fu, W.; Loh, T.-P. Direct Substitution of Secondary and Tertiary Alcohols To Generate Sulfones under Catalyst- and Additive-Free Conditions. *Org. Lett.* **2018**, *20* (17), 5353−5356.
(19) Blieck, R.; Lemouzy, S.; Taillefer, M.; Monnier, F. Regio-, Stereo- and Enantioselective α-Addition of Carbonyl Nucleophiles to Allenamides Catalyzed by a Synergistic Copper/Enamine System. *ChemRxiv*, April 15, **2020**, ver. 3. DOI: 10.26434/chemrxiv.11830935.v3.
(20) Abed Ali Abdine, R.; Pagès, L.; Taillefer, M.; Monnier, F. Hydroarylation of N-Allenyl Derivatives Catalyzed by Copper. *Eur. J. Org. Chem.* **2020**, *2020* (48), 7466−7469.
(21) Blieck, R.; Abed Ali Abdine, R.; Taillefer, M.; Monnier, F. Regio- and Stereoselective Copper-Catalyzed Allylation of 1,3-Dicarbonyl Compounds with Terminal Allenes. *Org. Lett.* **2018**, *20* (8), 2232−2235.
(22) Blieck, R.; Bahri, J.; Taillefer, M.; Monnier, F. Copper-Catalyzed Hydroamination of Terminal Allenes. *Org. Lett.* **2016**, *18* (6), 1482−1485.
(23) Perego, L. A.; Blieck, R.; Groué, A.; Monnier, F.; Taillefer, M.; Ciofini, I.; Grimaud, L. Copper-Catalyzed Hydroamination of Allenes: From Mechanistic Understanding to Methodology Development. *ACS Catal.* **2017**, *7* (7), 4253−4264.
(24) Perego, L. A.; Blieck, R.; Michel, J.; Ciofini, I.; Grimaud, L.; Taillefer, M.; Monnier, F. Copper-Catalyzed Hydroamination of N-Allenylazoles: Access to Amino-Substituted N-Vinylazoles. *Adv. Synth. Catal.* **2017**, *359* (24), 4388−4392.
(25) Blieck, R.; Perego, L. A.; Ciofini, I.; Grimaud, L.; Taillefer, M.; Monnier, F. Copper-Catalysed Hydroamination of N-Allenylsulfonamides: The Key Role of Ancillary Coordinating Groups. *Synthesis* **2019**, *51* (5), 1225−1234.
(26) Blieck, R.; Taillefer, M.; Monnier, F. Copper-Catalyzed Hydrocarboxylation of N-Allenyl Derivatives. *J. Org. Chem.* **2019**, *84* (17), 11247−11252.
(27) Lakhdar, S.; Tokuyasu, T.; Mayr, H. Electrophilic Reactivities of α,β-Unsaturated Iminium Ions. *Angew. Chem., Int. Ed.* **2008**, *47* (45), 8723−8726.

https://doi.org/10.1021/acs.joc.1c01345
J. Org. Chem. 2021, 86, 15695−15701